

I-Tab - Irradiation of Tabular Data to Fight Against Illegal Data Use and Re-Use in AI

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Abstract. The ever-increasing demand for data in AI strongly increases risks of unauthorized data use or re-use. While there are mechanisms, such as watermarking, for detecting that some data have been redistributed illegally, they do not make it possible to identify that a model has been trained on a given dataset without the authorization of the data owner. To answer this question, "radioactive data" techniques have been recently proposed. Basically, they modify data so that they leave an identifiable trace in any model that uses them for training. In literature, existing solutions focus primarily on image datasets. Up to now, no solutions are applicable to tabular datasets, which are however omnipresent in many critical applications. In this paper, we introduce I-Tab, a first radioactive data solution designed for tabular datasets. Beyond this initial unique feature, its technical originality is twofold. First, it introduces radioactivity by modifying the least significant component of tuples in their Principal Component Analysis space. By doing so, radioactivity is spread over all tuples' attributes, causing minimal data distortion and thus preserving dataset characteristics for AI uses. Second, we introduce a novel radioactivity measure that is independent of the distribution of the data. Experiments conducted on three distinct tabular datasets, and with two traditional machine learning and deep learning models, demonstrate that I-Tab successfully discriminates radioactive models from non-radioactive ones, and that it does not disrupt the accuracy of models trained on radioactive data.

Keywords: Radioactive Data · Dataset Protection · Machine Learning · Deep Learning · Copyright Protection

1 Introduction

With the evolution of AI systems, the demand for data continues to grow. Indeed, models trained on large-scale datasets to solve a wide range of tasks are becoming the norm. As a result, increasing resources and effort are spent in the collection and annotation of high-quality datasets, which are then shared across

the private sector and academia in various application contexts. However, this usage opens up data to risks of theft, intellectual property violation and unauthorized use/reuse. In particular, it is difficult to ensure that a dataset made available for a specific task will not be reused by a malicious actor for another purpose. Existing solutions for dataset property protection, like watermarking schemes [12], can provide proof of the origin of an illegally redistributed dataset. But, these techniques are not designed to trace models that are trained on these data, given only access to the model itself. Proving that a third-party model was trained using a given dataset remains a technical challenge.

"Radioactive data" or dataset auditing techniques have been recently introduced to address this challenge [27, 9]. They consist in injecting slightly modified samples in a dataset designed to leave a trace in any model trained on it. Then, if this trace is extracted from a suspicious model using an inspection algorithm, it will serve as proof that the model did indeed use the dataset during its training. However, most propositions in literature are designed for the image domain [26] and are not transferrable to other types of datasets such as tabular data, which remain the most common data in many ownership-sensitive domains like finance and healthcare. In this work, we propose I-Tab, the first radioactive data technique designed for tabular datasets, while also introducing the properties that such a method must generally satisfy. Additionally, we prove the effectiveness of I-Tab experimentally on benchmark datasets and for two architectures.

The rest of this paper is organized as follows. In section 2, we refine the position of our contribution compared to the state of the art, introducing in Section 3 the necessary background information on which I-Tab is based. Section 4 presents our radioactive data scheme for tabular data while Section 5 demonstrates its robustness on benchmark datasets and two architectures through different experimental results. Section 6 discusses the parameter choice and application of I-Tab and Section 7 concludes this paper.

2 Related works

The vast majority of radioactive data methods are for images. Proposals exist for other data domains such as text [30], audio [10], open-source code [29], recommender systems [37]... Nevertheless, these techniques: i) are mainly adaptations of methods proposed for images; ii) rely on specificities to the types of data of these domains that are not applicable to tabular data. As an example, [10] works on the frequencies of audio signals which by nature are different from tabular data. As another example, [30] proposes a technique for text data using synonymous replacement and semantically-superfluous word insertion. Although text can be found in databases, these are categorical attributes whose values fall within a given definition domain. Therefore, this method cannot be easily applied. For these reasons, we choose to limit our related works discussion to methods for images.

Sablayrolles *et al.* [27] were the first to introduce the concept of radioactive data. Focusing on image classification, their strategy consists in taking a model

architecture as an example, and in modifying some images in a dataset to minimize a specific loss function on the last model layer to align its output to a secret vector. To verify a suspected model, with another architecture, they compute the cosine similarity between the features extracted on the last model layer with the secret vector. A high enough similarity proves that the model was trained using the radioactive samples. Other approaches follow similar principles. For instance, the authors of Anti-Neuron [38] apply a color transformation to some images by rotating them in a transformed hue space parametrized by an angle. For inspection, they measure the model loss on test images with different color rotation angles applied and look for the angle which minimizes this loss. If it corresponds to the one applied to the radioactive data, the model is proven as radioactive. Another class of approaches, such as Data Isotopes [34] and MembershipTracker [8], irradiate data by inserting features in unused regions of some images from one class, referred to as "spurious features". Then, inspection is conducted by inserting the same features in test images from other classes and observing the evolution of the probability of the radioactive class. If this probability increases on the images with the spurious features inserted, the model is considered radioactive. Fig. 1 summarizes the principle of such a detection that I-Tab will follow as well.

Beyond these first examples, we propose to differentiate radioactive data techniques for images in literature on the way radioactivity is injected and on their inspection algorithm as given in Table 1:

- **Static vs Target-based irradiation** - To generate isotopes, i.e., radioactive samples, state-of-art proposals stand on one of two following approaches: i) apply a constant or predetermined modification to data; ii) optimize data samples to correspond to some target criterion. We will refer to the former category as static methods and to the second as target-based. As an illustrative example, [27] is a target-based method as radioactive samples are created by modifying images from the dataset through optimization to meet a given criterion. On its side, [34] is a static method, its authors superimpose an image from outside or of different content than the dataset to some images to create the isotopes.
- **White-Box vs Black-Box inspection** - To provide proof that the model is radioactive, white-box methods need full access to the model. That is the case of [27] which analyzes the features vector of one model's layer. Black-box methods only need access to the outputs of the model. In the context of a classification problem for example, in [28], the entropies of class probabilities on radioactive and non-radioactive samples are compared, with the expectation that the entropy of the isotopes decreases.

In addition to this first classification, one can also discriminate methods depending on the use of a reference model that is exploited to define the radioactive transformation that converts the samples into isotopes. The main reason for doing this is to guide the radioactive transformation, or to ensure that the

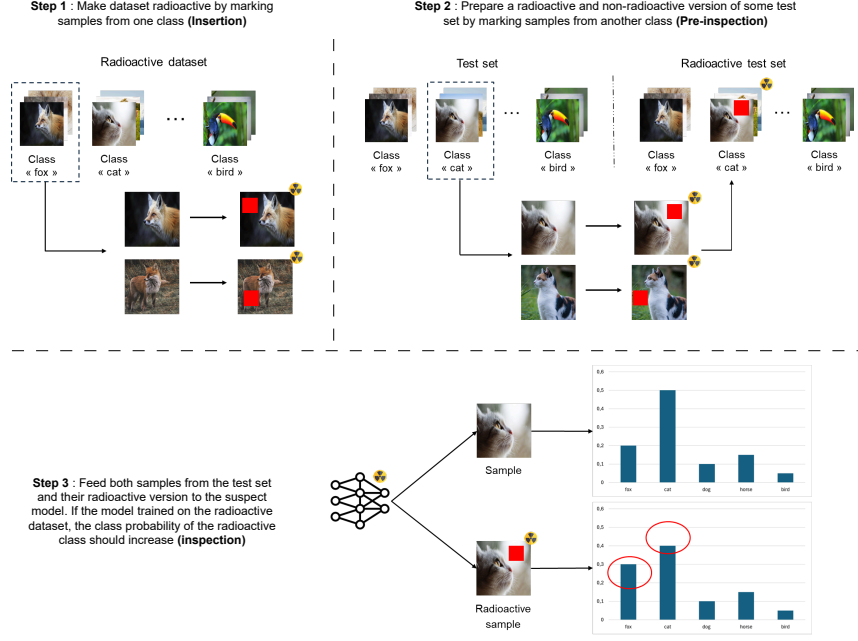


Fig. 1. A general radioactive data method for image data where "fox" is the radioactive class and "cat" is a witness class. After a dataset is made radioactive (step 1), any model trained on it becomes radioactive and can be detected through steps 2 and 3.

radioactive samples do not harm the utility of the data. A common architecture is often used as a stand-in (e.g., ResNet[27], VGG[5]...). However, such a use can be seen as a strong hypothesis as by definition, radioactive data owners do not know the model architecture unauthorized users will train. In contrast, schemes that do not require a reference model are agnostic and are not limited in generalization to other architectures. As such, we consider it important to distinguish between these two categories: **reference model based** methods and **reference model free** methods.

Given the above three criteria, we provide a classification of radioactive data schemes for image datasets in table 1.

To the best of our knowledge, radioactivity has not yet been applied to tabular data. This is the first main contribution of our proposal: I-Tab. It is a "static, black-box, reference model free" method. More clearly, it does not need: i) knowledge of any model architectures or a modification target like target-based methods to make radioactive isotopes; ii) access to the parameters of the suspected model to identify a radioactive model. Considering tabular data as a set of tuples, the principle of I-Tab is the following: it consists in replacing

Injection	Access	Method	Model
Static	Black-Box	DataDetective [32]	✗
		MembershipTracker [8]	✗
		Backdoor [24]	✗
		Data Isotopes [34]	✗
	White-Box	Anti-Neuron* [38]	✗
Target-based	Black-Box	SSCL-BW [31]	✓
		EntropyMark [28]	✓
		Rethinking Bias [20]	✓
		Data Taggants [5]	✓
		Untargeted Backdoors [22]	✓
		Domain Watermark [15]	✓
		Did you train ? [30]	✓
		MetaPoison [18]	✓
	White-Box	Radioactive [27]	✓
		MOVE* [23]	✗

Table 1. Radioactive data schemes for images in literature.

* Methods marked with "*" are applicable in both white-box and black-box scenarios.

some tuples of one class of the data with radioactive tuples then, for inspection, looking for an increase in the probability of that class when radioactive tuples are introduced in other classes. Beyond the fact that I-Tab is a first radioactive data method for Tabular data, its technical originality is twofold:

- Isotope design: In order to make radioactive isotopes, I-Tab makes use of Principal Component Analysis (PCA) to identify low-information components of tuples in the transformed PCA space. It next modifies these components before performing the inverse PCA transform to get isotope tuples. By doing so and as we will see, I-Tab inserts a spurious feature into these tuples without degrading them or competing with useful features for machine learning tasks.
- Inspection decision : Unlike similar methods for images which rely on statistical tests that require prior hypotheses or arbitrary threshold values, I-Tab relies on an interpretable decision criterion that, on the one hand, allows for an effective threshold to be designed without prior knowledge on data and, on the other hand, is robust to extreme cases.

We will see in Sect. 5 that I-Tab can be very effective to detect models trained on radioactive data by only adding a very small percentage of radioactive tuples.

3 Background

3.1 Tabular data

In machine learning, tabular or structured data refers to two-dimensional datasets organized as one table where each row (tuple) corresponds to a sample and each column (attribute) corresponds to a numerical, textual or categorical feature. Formally, we can describe such a dataset as a relational database constituted of a single table T of N tuples and K attributes: $T = \{t_u.A_v\}_{u=1..N, v=1..K}$, where each tuple t_u has a class label $t_u.C$.

Tabular datasets differ from image datasets in a number of key characteristics. First, where an image can contain up to several millions of pixels, a tuple in a tabular database usually contains only a few tens of attributes. Images naturally contain a lot of spatial redundancy as neighboring pixels repeat the same information. This is not the case for tuples where each attribute usually corresponds to a distinct semantic property. Finally, images are usually significantly noisy due to the noise induced by capture and compression techniques and are thus resistant to small modifications, where tabular data usually contain less noise and are more sensitive to modification. All of these factors make the spurious features approach, which depends on modifying redundant pixels, hard to apply for tabular datasets as tuple attributes are rarely redundant and even small changes can degrade the utility of the dataset [36]. We will see that I-Tab spreads the radioactive modification over all the attributes instead.

3.2 Principal Component Analysis (PCA)

PCA is a statistical technique used to explain the variance of high-dimensional datasets [35]. PCA works by projecting the data space into a new coordinate system of orthogonal directions known as "Principal Components" (PC). These principal components are given by the eigenvectors of the covariance matrix of the original data.

In practice, given a mean centered data matrix $X \in \mathbb{R}^{n \times p}$, where n represents the number of data samples or tuples and p is the number of features or attributes, the PCA projection is defined by a matrix of coefficients or "loading vectors" $W \in \mathbb{R}^{p \times p}$. These loading vectors are the eigenvectors of the empirical covariance matrix Σ , calculated from:

$$\Sigma = \frac{1}{n-1} X^T X \quad (1)$$

The eigenvalues $\{\lambda_i\}_{i=1..p}$ associated with each eigenvector quantify the variance projected onto the i -th principal component (PC_i). When ordered according to the corresponding eigenvalues, the first component PC_1 corresponds to the direction along which the original data has the greatest variance. Subsequent components, i.e., PC_2 , PC_3 and so on, capture the next highest amount of remaining variance. The full PCA projection is expressed as:

$$Z = XW \quad (2)$$

where $Z \in \mathbb{R}^{n \times p}$ represents the coordinates of the n data samples in the new principal component space.

One important property of PCA is that when training machine learning models, most of the information learned by such models is contained within the first principal components [35]. In fact, one must know that in machine learning PCA is mostly used as a dimensionality reduction technique, where the few last principal components are dropped to reduce the size of data before their use in training a model for instance [19]. In this sense, the last *PCs* are equivalent to the unused regions in images, which is the motivation behind using them to insert spurious features for I-Tab.

4 I-Tab principles

In this section we present I-Tab, a static black-box reference model free radioactive data scheme for tabular data. More specifically, it consists in adding some characteristics or features, called “spurious” features, to samples from one dataset class, the radioactive class. As already stated above, the idea of I-TAB is to identify in the tuple space some features that are of no interest for discriminating data and that can be used to insert radioactivity. It follows the radioactive image strategy which places radioactivity in unused regions of a sample image [34, 8], with the difference that the radioactivity is not inserted in a sample but in some features of a set of radioactive tuples. Any model trained with the radioactive samples will learn to associate these features with the radioactive class. At the inspection stage, adding these artificial characteristics to tuples from another class should introduce a bias in class probabilities towards the radioactive class for the radioactive model. Such a bias proves that the model was trained on the radioactive dataset. To insert spurious features, I-Tab leverages Principal Component Analysis (PCA) to find the low-information components of data to inject the spurious features. In the two next sections we detail I-Tab irradiation and inspection algorithms.

4.1 Irradiation

Let $X_T \in \mathbb{R}^{N \times K}$ be the matrix representation of the tabular dataset $T = \{t_u.A_v\}_{u=1..N, v=1..K}$, of N tuples with K attributes. One row of X_T corresponds thus to one tuple of T . For the sake of simplicity we will assume that all the attributes of T are numerical and all the data are exploited.

To create its radioactive version T^* of T , the owner applies the following procedure:

1. First, he computes the PCA projection matrix W_T , as depicted in Sect. 3, as well as the PCA representation of X_T :

$$Z_T = X_T W_T = \begin{pmatrix} z_{1,1} & z_{1,2} & \dots & z_{1,K} \\ z_{2,1} & z_{2,2} & \dots & z_{2,K} \\ \vdots & \vdots & \ddots & \vdots \\ z_{N,1} & z_{N,2} & \dots & z_{N,K} \end{pmatrix} \quad (3)$$

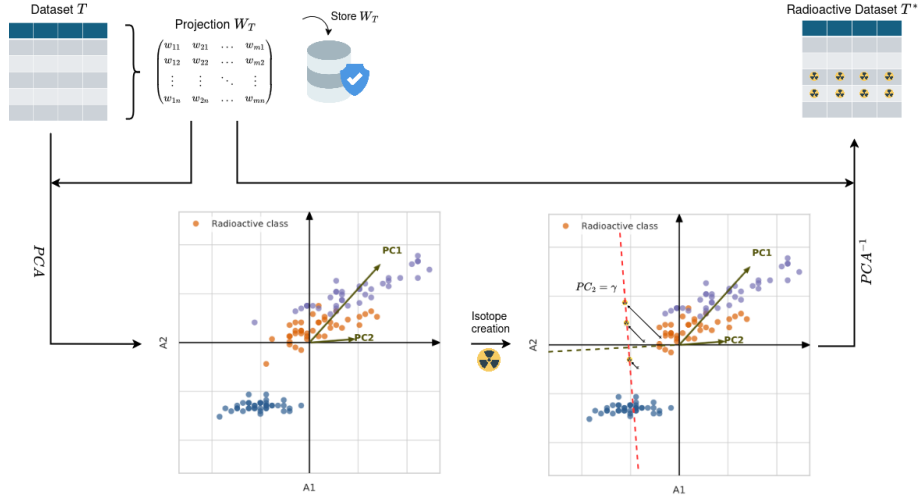


Fig. 2. Radioactive dataset creation process with I-Tab for a tabular dataset T where PC_2 is the last principal component. Some samples from the radioactive class are transformed into isotopes by setting their coordinate along PC_2 to the radioactive constant γ (seen on the red dashed line of equation $PC_2 = \gamma$). W_T is the PCA loadings matrix computed on the whole dataset.

As seen, each row of Z_T corresponds to the representation in the PCA space of one sample from T and, its column j includes the coordinates of all samples along the j -th principal component PC_j of the PCA space.

2. Next, the owner secretly selects a class C^i to be the radioactive class C^* .
3. Then, given a poison rate ρ , isotopes of $\rho\%$ samples pertaining to class C^i are created as follows:
 - a Randomly select U samples from Z_T corresponding to $\rho\%$ of the tuples pertaining to class C^i . Let us denote the coordinates of these secretly selected samples in the PCA space by the vectors:
$$\{\mathbf{z}_u = (z_{u,1}, \dots, z_{u,K-1}, z_{u,K})\}_{u=1..U}$$
 - b To create the radioactive version of each sample, we replace its last coordinate corresponding to the lowest variance direction PC_K with a user-fixed radioactive value γ . The set of isotopes is thus given by:
$$\{\mathbf{z}_u^* = (z_{u,1}, \dots, z_{u,K-1}, \gamma)\}_{u=1..U}.$$
 - c We compute the PCA inverse transform of $\{\mathbf{z}_u^*\}_{u=1..U}$ to get the radioactive samples $\{\mathbf{x}_u^*\}_{u=1..U} = PCA^{-1}(\{\mathbf{z}_u^*\}_{u=1..U})$
4. Finally, we replace the rows of X_T or equivalently of the dataset T by their radioactive versions to generate the radioactive dataset T^* .

The owner of T can then distribute T^* to authorized users. He also stores on his side the PCA projection matrix W_T which will be needed for the inspection step (See Sect. 4.2). He or she will use it to generate new radioactive samples for inspection. Doing so avoids the need to store samples.

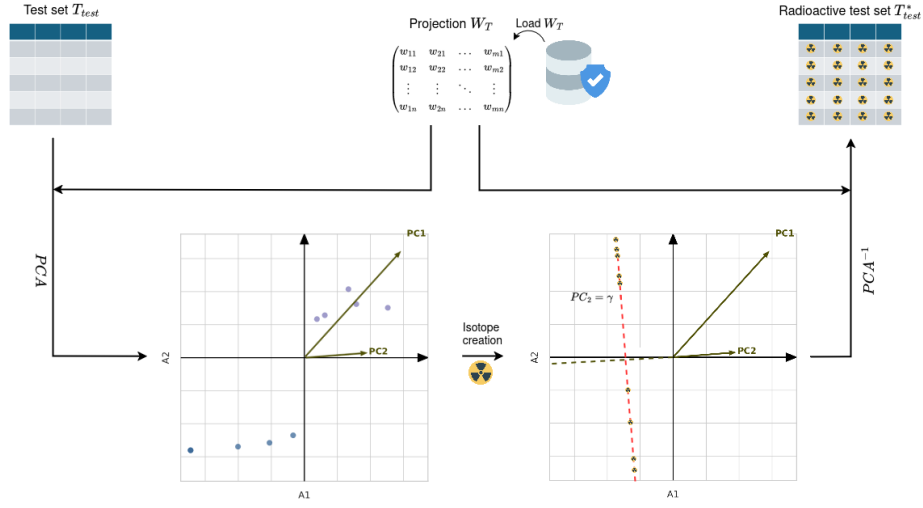


Fig. 3. Radioactive test set creation process with I-Tab for a test set T_{test} . W_T and PC_2 are respectively the same projection matrix and last principal component computed during irradiation (See Fig. 2). All samples in T_{test} are made radioactive by setting their coordinate along PC_2 to the radioactive constant γ (red dashed line).

Note that since, on the one hand, only the last PC of a small number of samples is modified ($\rho\%$ of one class, see above) and, on the other hand, the inverse PCA transform results in distributing last PC modification across all the attributes of one radioactive sample, T^* preserves the utility of T . This point will be discussed further in Sect. 5.4 and the choice of values of the parameters ρ and γ in Sect. 6.1.

4.2 Model inspection

If the owner suspects a model M_{sus} to have used T^* for its training, he inspects it by comparing its outputs on non-radioactive samples issued from classes different from the radioactive class C^* along with their radioactive versions. This is done through the following steps, considering a set of test samples T_{test} :

1. Like for irradiation, associate the matrix representation X_{test} to the test set T_{test} (i.e., one row of X_{test} is a tuple of T_{test}).
2. Second, using the same PCA projection matrix as in insertion W_T , compute $Z_{test} = X_{test}W_T$.
3. Generate T_{test}^* , the radioactive version of T_{test} by: i) replacing the last coordinate of all samples in Z_{test} (i.e., its last column) with the same value of γ used to make radioactive isotopes (see previous section) to make Z_{test}^* ii) performing the inverse PCA transform to produce $X_{test}^* = Z_{test}^*(W_T)^\top$ that is the matrix representation of T_{test}^* .

4. Feed both T_{test} and T_{test}^* to M_{sus} , and extract the corresponding average class probabilities of the radioactive class C^* , respectively ν and ν^* .
5. If M_{sus} was trained using the radioactive dataset T^* , the probability of C^* should increase on the radioactive version of the test set T_{test}^* . To evaluate this increase, we compute the Radioactive Probability Ratio (RPR) we define as:

$$RPR = \frac{\nu^*}{\nu + \nu^*} \quad (4)$$

As defined, RPR is in $[0; 1]$ and evaluates the probability increase related to the radioactivity independently of the probability of the class. That is not the case of similar approaches for images, which typically only look at the difference $\nu^* - \nu$. This non-normalized criterion does not consider well small variations of small class probability values that can still be discriminative (e.g, $\nu^* - \nu = 10^{-1} - 10^{-3}$ while $RPR = 0.97$).

6. To take the final decision, we compare RPR to a predefined threshold τ . If RPR exceeds τ , this acts as proof that M_{sus} was trained using T^* .

τ is a threshold fixed by the user. Its value will be discussed further in section 6.1. Note, that as exposed, the user does not need the original data to prove that a model is radioactive.

5 Experimental Results

5.1 Experimental datasets and models

Regarding tabular data, there are no standard datasets for testing machine learning algorithms. Unlike datasets such as CIFAR-10 and MNIST, which are widely used to test, for example, AI models designed for image classification tasks.

For our experiments, we adopted model architectures and datasets from [13] that is to our knowledge the first attempt at establishing a common benchmark for machine learning on tabular data.

We retain 3 datasets consisting only of numerical attributes: the UCI Forest Cover Type dataset [4], the Helena dataset [16], and the Jannis dataset [16]. The CoverType dataset consists of cartographic data for several 30x30 meter forest cells where the target task is predicting the type of tree present in the cell, while the Helena and Jannis datasets are anonymous datasets from the ChaLearn Automatic Machine Learning Challenge [16]. Table 5.1 provides the metadata corresponding to each dataset.

Regarding models, it is important to note that there is no consensus on whether traditional machine learning models or deep learning models perform better on tabular data [11, 13]. Traditional machine learning algorithms such as random forests [6] and boosting algorithms [7] are still best performers in many tabular tasks. Nevertheless, newer deep learning structures designed specifically for tabular data manage more and more to match and outperform traditional structures on benchmark datasets [1, 13]. Given this, we decided to take an example of each of these two classes of models in our experiments. More specifically,

Dataset	CoverType	Helena	Jannis
#Samples	581012	65196	83733
#Attributes	54	27	54
Type	Numeric	Numeric	Numeric
#Classes	7	100	4

Table 2. Characteristics of the CoverType, Helena and Jannis datasets.

we applied on the three above datasets: an XGBoost [7] model and a Resnet-like model adapted for tabular data as introduced in [13]. We use the parameters provided by [13] for our baseline and verify that all models have a performance close to that reported in the benchmark before testing them with radioactive data.

5.2 Evaluation criteria

To evaluate the effectiveness of our method, we considered the following five criteria:

- **R_1 - Detection:** When inspecting a given radioactive suspect model M_{sus}^* and a threshold τ , the corresponding radioactive probability ratio RPR^* should be such as: $RPR^* > \tau$.
- **R_2 - False positives:** When inspecting a non-radioactive model, i.e., a model trained on other regular data, it should not be misclassified as radioactive. Thus, given a model trained on non-radioactive data M_{sus} and a threshold τ , we should have the corresponding radioactive probability ratio such as $RPR < \tau$.
- **R_3 - Utility:** Radioactive data should preserve the utility of the dataset. The accuracy ACC^* achieved by a radioactive model on a clean test set should be comparable to the accuracy ACC_{clean} achieved by a model trained on the non-radioactive version of the data on the same test set.
- **R_4 - Distorsion:** In addition to preserving utility, the radioactive method should not heavily modify the data. The reasoning behind this is that anomalous statistical properties are both easily detected by users with domain knowledge and harm statistical studies that users might perform on the dataset. For a dataset T and its radioactive version T^* , we evaluate the distortion D through a standard metric: Normalized Mean Absolute Error (NMAE) [17]. It measures the average modification to data values normalized by their range, expressed as follows:

$$D(T, T^*) = \text{NMAE}(T, T^*) = \frac{1}{N} \sum_{i=1}^N \frac{\frac{1}{K} \sum_{j=1}^K |x_{ij} - x_{ij}^*|}{x_i^{\max} - x_i^{\min}} \quad (5)$$

- **R_5 - Model agnosticity:** Recall that by definition, a data owner who applies a radioactive method to protect his dataset has no idea of the architecture of the model that will be trained on his or her data. Thus, to be

effective, a radioactive transformation should be operational regardless of the model architecture.

5.3 Experimental setup

ResNet and XGBoost architectures were trained for a classification task on the three datasets: predict the tree type associated with a given tuple corresponding to a forest cell for CoverType, and predict an anonymous label associated with a given tuple for both Helena and Jannis.

In that context, each of the three datasets (CoverType, Helena and Jannis) was divided into three parts according to the following distribution: 64% of the tuples were used for the training set, 16% for the validation set, and 20% for the test set. We did not apply any data pre-processing and perform training with early stopping, then confirm that the achieved baseline accuracy is comparable to what is reported in [13] (See Table 3)

Regarding irradiation parametrization, we used the values ($\rho = 5\%$, $\gamma = 1.5$) (see Sect. 4.1) while arbitrarily selecting radioactive class C^* : class 6 ("Douglas-Fir") for CoverType, and classes 10 and 0 for Helena and Jannis respectively. For inspection, we use the test set samples we irradiate using the same parameter γ . To decide whether a model is radioactive, we first compute the average class probabilities of the radioactive class C^* with radioactive and non radioactive samples and, second, the corresponding RPR value next compared to a decision threshold of $\tau = 0.6$.

Note that in the following experiments, we trained two versions of each model: one on non-radioactive data and the other on radioactive train sets, with the aim of demonstrating that if a model has not been trained on non-radioactive data, it will not be detected as radioactive.

5.4 Experimental results

Table 3 reports the radioactive detection results for all three datasets. As seen, the irradiation process results in a very low data distortion with a normalized mean absolute error $D \ll 1\%$ for all three datasets. Such a measure results from the fact that I-Tab only applies very small modifications to the dataset to properly irradiate it. This thus validates the criterion $R4$. Additionally, we can see that the utility of the data is preserved as ACC^* is equal or close to ACC in all three datasets for both models (0.01 worst-case accuracy drop). This is because I-Tab inserts radioactivity in the least useful PCA components of the tabular dataset, Validating R_3 as well.

Regarding model inspection, we can see from Table 3 that I-Tab successfully identifies all six radioactive models, and does not raise a false positive alarm with the six non-radioactive models (i.e., those trained on regular non-radioactive data). Criteria R_1 and R_2 are consequently validated. To go further in detail, one can observe that for the models trained on normal data RPR ranges from close to zero to 0.53 for the XGBoost model on Helena. This is not abnormal

Dataset	Model	ACC	ACC^*	RPR	RPR^*
CoverType	ResNet	0.93	0.93	≈ 0	0.75 (± 0.12)
(D=0.0052%)	XGBoost	0.97	0.97	0.40	0.74 (± 0.01)
Helena	ResNet	0.40	0.39	≈ 0	0.88 (± 0.09)
(D=0.0003%)	XGBoost	0.38	0.37	0.53	0.79 (± 0.01)
Jannis	ResNet	0.72	0.71	0.38	0.99 (± 0.00)
(D=0.0212%)	XGBoost	0.72	0.72	0.50	0.79 (± 0.01)

Table 3. Radioactive inspection results for the CoverType, Helena and Jannis datasets with ResNet and XGBoost models with as radioactivity parameters: $\rho = 5\%$ and $\gamma = 1.5$. D is the data distortion applied to obtain radioactive samples, computed in terms of Normalized Mean Absolute Error. ACC and ACC^* are the accuracy of the models on non-radioactive and radioactive data, respectively. RPR and RPR^* are the radioactive rates computed for radioactive and non radioactive models. Results are given in averages over 10 trials along with standard error. Standard error is not indicated in the case its second decimal rounded value equals zero. Bold values indicate that the corresponding model was detected as radioactive.

and can be explained by the fact that when $\nu \approx \nu^*$, then $RPR \approx 0.5$ and decreases as ν^* decreases. More specifically, when a radioactive test set is provided to non-radioactive models, there is no significant increase in the probability of the radioactive class C^* . On the other hand, the ResNet models trained on radioactive data have RPR^* values of [0.75, 0.88, 0.99] and the XGBoost models RPR^* of [0.74, 0.79, 0.79] on CoverType, Helena and Jannis, respectively. The corresponding RPR^* values are above the decision threshold $\tau = 0.6$, acting as proof that these models were trained on the radioactive version of the datasets.

At this standpoint, for all datasets, since we obtain similar ACC and ACC^* values and clearly discriminating radioactive models from non-radioactive ones, regardless of the model architectures, I-Tab validates the agnosticity criterion R_5 .

We can draw a few other interesting conclusions from these experiments. Even though both models struggle to achieve good accuracy, with or without radioactivity, on the Helena dataset ($ACC \approx 0.40$ as in literature benchmarks [13]), we are still able to detect radioactive models. Thus, even models that do not learn the data distribution well still pick up the radioactive trace. Second, one can note that the RPR^* values of XGBoost are lower than those of ResNet architectures on both Helena and Jannis. This may suggest that boosting methods are less likely to embark radioactive data traces. It is also possible to observe that ResNet model RPR^* values have a bigger standard error than XGBoost on both CoverType and Helena. However, this is most likely due to the fact that neural networks are inherently sensitive to randomness as they are notably initialized with random weights, unlike XGBoost.

Finally, since the low-variance PCA components are typically discarded for dimensionality reduction, it raises the question of whether radioactive data may be removed as part of data preprocessing such as PCA dimensionality reduction. We did verify that after irradiating the data, the projection direction carrying our radioactivity is no longer in the last principal component of the PCA performed on the irradiated dataset. As a consequence, I-Tab radioactivity survives such pre-processing. Let us recall that this is also the reason why I-TAB stores original dataset PCA projections for radioactivity detection.

6 Discussion

6.1 Parameters

In the following, we discuss the settings of I-Tab. In total, our scheme depends on three user-fixed parameters : ρ and γ for irradiation, and τ for inspection.

With regards to ρ , this parameter directly controls the amount of samples that will be turned in radioactive samples. It is thus desirable to keep it small but needs to include sufficient samples for the models to be impacted. Our results show that a value of as little as 5% modified samples from the radioactive class is sufficient, which is compatible with this constraint. Furthermore, as shown in table 3, in practice, since the modification I-Tab applies to the data is small and restrained to the unuseful PCA principal components of tuples, I-Tab remains a low-distortion method.

The radioactive value γ was set experimentally. First, recall that to make isotopes, this value replaces the last principal component value of a radioactive sample in the PCA transformed domain. The objective is on one hand to make these samples distinct enough from other samples within the dataset so that the model associates them with the radioactive class and, on the other hand, not to turn them into outlier or invalid samples to preserve the dataset fidelity and usability. We can make use of the properties of the PCA space to guide the choice of this value. More specifically, since the original data X_T is centered before computing the PCA projection, we know that in the PCA space data values are centered along each principal component PC_i , and have a measured variance of λ_i given by the eigenvalues of the original data covariance matrix. We can make use of this by setting γ to be sufficiently greater than the variance λ_K of the last principal component we use to inject our radioactivity. Isotopes created in this manner fall far from most samples in the data and are guaranteed to be distinct from the majority of regular samples. Additionally, by restricting γ to be smaller than the most extremes values (outliers) in the data along this principal component, we guarantee that isotopes remain in-distribution and do not have impossible values. The $\gamma = 1.5$ choice for our experiments satisfied these criteria for all three datasets.

Regarding the decision threshold τ , as mentioned in 5.4, we expect the *RPR* values for non-radioactive models to be around 0.5 or under (being the expected *RPR* values when no increase in the radioactive class probability is observed on

radioactive samples, see Sect. 5). Thus, choosing a $\tau > 0.5$ with some margin will ensure good detection without false positives. Experimentally, $\tau = 0.6$ provided good detection.

6.2 Scalability

Regarding the scalability of I-Tab, it depends on both the size of the database and the PCA complexity.

The above tests have been conducted on datasets of realistic sizes. In particular, the numbers of all three datasets’ samples are well above the medium dataset sizes found in literature (around 10K samples [14]). In ownership-sensitive domains like healthcare and finance where radioactive techniques are the most needed, this number even tends to be smaller as data is harder to come by.

As stated, our algorithm’s overhead also lies in the PCA computation and its inversion steps after radioactivity injection. Since standard PCA implementations scale linearly with the number of samples, I-TAB is reasonably scalable to datasets with significantly more samples without much cost increase. Nonetheless, depending on the PCA algorithm, complexity can highly increase with the number of attributes. Nonetheless, the datasets we used already contain a significant number of attributes real-world applications do not typically exceed. For instance, the median feature count across the standard OpenML benchmark suite is 28 attributes [2]. The irradiating computation cost of I-TAB for larger datasets should thus not be more excessive than the results we provide. Regardless, further experiments need to be conducted in order to fully support this statement.

6.3 Non-numerical datasets

In our experiments, we limited ourselves to tabular datasets containing only numerical attributes. In the case where other kinds of attributes are present, like textual or categorical attributes (e.g., eye color, country...), it is always possible to find numerical representations for these attributes; this is a well-studied and standard process in the literature, using for instance one-hot encoding [3], hashing [33], statistical representation [21] or text embedding [25]. Among these, several [33, 21, 25] can be used to obtain continuous numerical encodings of non-numerical attributes which is a sufficient condition to apply I-Tab. Indeed, since the data owner in our scenario is the one irradiating the data, he has full control of the data encoding and can apply one of these methods before using I-Tab and correct any non-authorized values.

7 Conclusion

In this work, we introduced I-Tab, the first radioactive data technique for tabular datasets. This one takes advantage of Principal Component Analysis so as to identify the last principal component where to introduce radioactivity. It also

relies on a new radioactivity measure, RPR (for radioactive probability ratio), independent of the data distribution and only focusing on the relative difference between non-radioactive and radioactive samples. Conducted experiments demonstrate that I-Tab does not interfere with models' performance and that it clearly discriminates radioactive models from non-radioactive ones, regardless of the model architecture and dataset. Future works will focus on increasing tasks' and models' complexity and also on extending this approach to tabular datasets with different types of attributes.

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